Yumeng He

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RESEARCH INTERESTS

My research interests are computer vision, graphics, and robotics, with a focus on physical AI, object/scene generation, articulated and kinematic asset modeling, physics-based simulation, real-to-sim pipelines, and policy learning.

EDUCATION

University of Southern California Aug 2024 – May 2026
Master of Science with Thesis *Los Angeles, CA, USA*

- Cumulative GPA: 3.88/4.0
- Computer Science major
- Advisor: Prof. Jernej Barbič

University of Toronto St. George Campus Sep 2019 – May 2024
Honours Bachelor of Science with High Distinction *Toronto, ON, Canada*

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major

PUBLICATIONS AND MANUSCRIPTS

[1] **VoroLight: Learning Quality Volumetric Voronoi Meshes from General Inputs**
Jiayin Lu*, Ying Jiang*, **Yumeng He***, Yin Yang, Chenfanfu Jiang
arXiv, 2025

[2] **SPARK: Sim-ready Part-level Articulated Reconstruction with VLM Knowledge**
Yumeng He*, Ying Jiang*, Jiayin Lu*, Yin Yang, Chenfanfu Jiang
arXiv, 2025

[3] **Birth of a Painting: Differentiable Brushstroke Reconstruction**
Ying Jiang*, Jiayin Lu*, Yunuo Chen*, **Yumeng He**, Kui Wu, Yin Yang, Chenfanfu Jiang
arXiv, 2025

[4] **Self-supervised dual-layer 2D normalizing flow method for industrial anomaly detection**
Zhenlian Miao, Guangzhu Chen, Xiaojuan Liao, Jiu Dai, **Yumeng He**
Applied Soft Computing (ASOC), 2024.

RESEARCH EXPERIENCE

USC Graphics | Master Thesis Aug 2025 – present
Supervisor: Prof. Jernej Barbič *Los Angeles, CA, USA*

- Enhancing traditional Kirchhoff–Love shell models by introducing a novel elastic energy formulation that accurately captures stretching, compression, and bending, while overcoming KL shells' inability to reproduce Poisson effects

UCLA AIVC Lab | Visiting student Jun 2025 – present
Supervisor: Prof. Chenfanfu Jiang *Los Angeles, CA, USA*

- Leading research on end-to-end articulation-aware 3D mesh generation, designing a pipeline that synthesizes multi-part objects from text or image conditions while mitigating over-segmentation via per-part image–guided local attention
- Performing joint optimization post-generation to infer plausible URDFs, enabling deployment in embodied simulation environments and supporting per-part texturing for visual realism and downstream manipulation compatibility

USC RESL Lab | Research Assistant Jul 2025 – present
Supervisor: Prof. Gaurav S. Sukhatme *Los Angeles, CA, USA*

- Designed a few-shot robotic manipulation task that uses Model Predictive Control to optimize material parameters for physically accurate simulation
- Led the Real2Sim pipeline for this project, building a digital twin from a single RGB D image of a cluttered tabletop scene with object segmentation, mesh reconstruction, and pose, scale, and placement estimation, and importing the reconstructed scene into ManiSkill3 for simulation

CDUT | Research Assistant May 2023 - Aug 2023
Supervisor: Prof. Guangzhu Chen *Remote*

- Faced with unlabeled industrial defects on MVTec AD and tasked to raise unsupervised detection/localization accuracy, proposed SS-DualFlow: a dual-layer 2D normalizing-flow that maps features to a Gaussian base to curb information loss and inserts an Exponential Space Attention module to focus on anomaly-salient regions. Result: image-level AUROC = 99.38% and pixel-level AUROC = 98.38% on MVTec AD; co-authored the Applied Soft Computing (2024) paper.

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL — Spring 2025

- Implemented and benchmarked four 2D incompressible fluid solvers, including Stable Fluids, Smoothed Particle Hydrodynamics (SPH), Particle In Cell (PIC), and Affine Particle In Cell (APIC) on identical scenarios, reporting performance-vs-accuracy trade-offs.

Collision Detection with Penalty Method and IPC | C++ — Spring 2025

- Built a physically accurate 3D jello cube simulator using a mass-spring system with structural, shear, and bend springs to model real-world deformation under force
- Implemented penalty-based collision detection to handle interactions with static obstacles, including inclined planes and spheres, ensuring realistic response under contact
- Upgraded basic collision logic with Incremental Potential Contact (IPC), reducing interpenetration to below $1e-4$ relative gap across $1,000+$ simulation steps, effectively achieving zero visible penetration and stable behavior

Inverse Kinematics with Skinning | C++ — Spring 2025

- Implemented Linear Blend Skinning and Dual Quaternion Skinning for 3D character deformation with smooth, realistic joint articulation
- Built an end-to-end FK–IK pipeline, supporting forward kinematics for pose propagation and inverse kinematics under joint constraints
- Developed and compared multiple IK solvers, including pseudoinverse least squares and damped least squares via Tikhonov regularization, improving numerical stability

Motion Capture Interpolation | C++ — Spring 2025

- Implemented four interpolation schemes (incl. linear & Bézier) with both Euler- and quaternion-based rotations to reconstruct and smooth optical mocap, mitigating gimbal lock and interpolation artifacts
- Generated natural, continuous transitions across keyframes for high-dimensional skeletal animation

Ray Tracing | C++, Rendering — Fall 2024

- Built a modular ray tracer with Phong shading, shadow rays, and recursive reflections, supporting both spheres and triangle meshes
- Added supersampling, soft shadows via area lights, and Möller–Trumbore intersection, achieving realistic visuals while balancing quality and performance

INDUSTRY EXPERIENCE

Software Developer Intern | Full Time — Aug 2022 – Aug 2023
HCL Canada Inc. — *Toronto, ON, Canada*

COMMUNITY SERVICES

- Member of Society of Women Engineers (SWE) — 2025 - 2026
- Mentor for Viterbi Graduate Mentorship Program — Fall 2025
- Mentor for Women in Engineering Mentorship — Fall 2025
- Mentor for Viterbi Graduate Mentorship Program — Summer 2025

INVITED TALKS

- *MPC Is All You Need.* [Co-presented]. USC SLURM Lab, hosted by Prof. Daniel Seita — Aug 2025

AWARDS & HONORS

- Dean's List Scholar — 2020 - 2024
- Ranked in the top 25% of contestants in the Galois Contest — 2017
- Ranked in the top 25% of contestants in the Cayley Contest — 2017
- Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics — 2016

TECHNICAL SKILLS

- **Programming Tools**: Linux, Windows, MacOS, C++, Python, HTML, CSS, JavaScript, MySQL, R, OpenGL, GLSL, Eigen, Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions